DRISHTI AI HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –1**
- **Problem Statement Title- AI-Based Human Behaviour Monitoring Using Pose Estimation and Gesture Analysis**
- **Team ID- DI_111**
- **Team Name (As Registered)- CaseBlitz**

PROPOSED SOLUTION: AUTOMATED AI INVIGILATION SYSTEM

- **AI-Powered Physical Hall Invigilator:** A real-time, GPU-free system designed for physical exam halls that monitors all students simultaneously using computer vision.
- **Optimized for Low-Power Hardware:** The prototype is built to run on affordable microcomputers like the Raspberry Pi 5 without requiring expensive GPU acceleration.

Detailed Explanation of the Proposed Solution

Detection and Tracking (YOLOv8 & Byte Track)
- Identifies students and prohibited items in real-time, assigning persistent IDs to ensure continuous monitoring even if students are temporarily occluded.

Dual-Triangle Pose Estimation
- Uses five facial keypoints to construct two geometric triangles, calculating head angles via the Law of Cosines to detect looking left, right, or back.

Dynamic Thresholding
- Employs individualized thresholds for each student to account for specific seating angles, reducing false alarms during natural movements like stretching.

Automated Cloud Logging
- Flags suspicious behavior and automatically saves timestamps and photographic evidence to an Excel file on Google Drive for administrative review.

How it Addresses the Problem

Overcoming Human Limitations
- Unlike manual invigilators who face fatigue and bias, this system provides objective, continuous, 300-degree surveillance.

Scalability for Large Physical Halls
- While most AI solutions are for online webcams, this system allows a single camera to monitor dozens of students in a large physical space.

Objective Evidence Collection
- Solves the lack of tangible proof in traditional invigilation by generating timestamped logs and photographic evidence for every incident.

Cost-Effective Deployment
- Eilminates the need for high-end GPUs, allowing institutions to deploy the system using off-the-shelf, affordable microcomputers.

Innovation and Uniqueness of the Solution

Novel Dual-Triangle Geometric Approach
- The first to measure head rotation through two distinct geometric triangles (nose-eyes and nose-ears) for granular accuracy.

GPU-Free Real-Time Efficiency
- Achieves 6–10 FPS on CPU-only hardware like Raspberry Pi, representing a significant leap in behavioral detection efficiency.

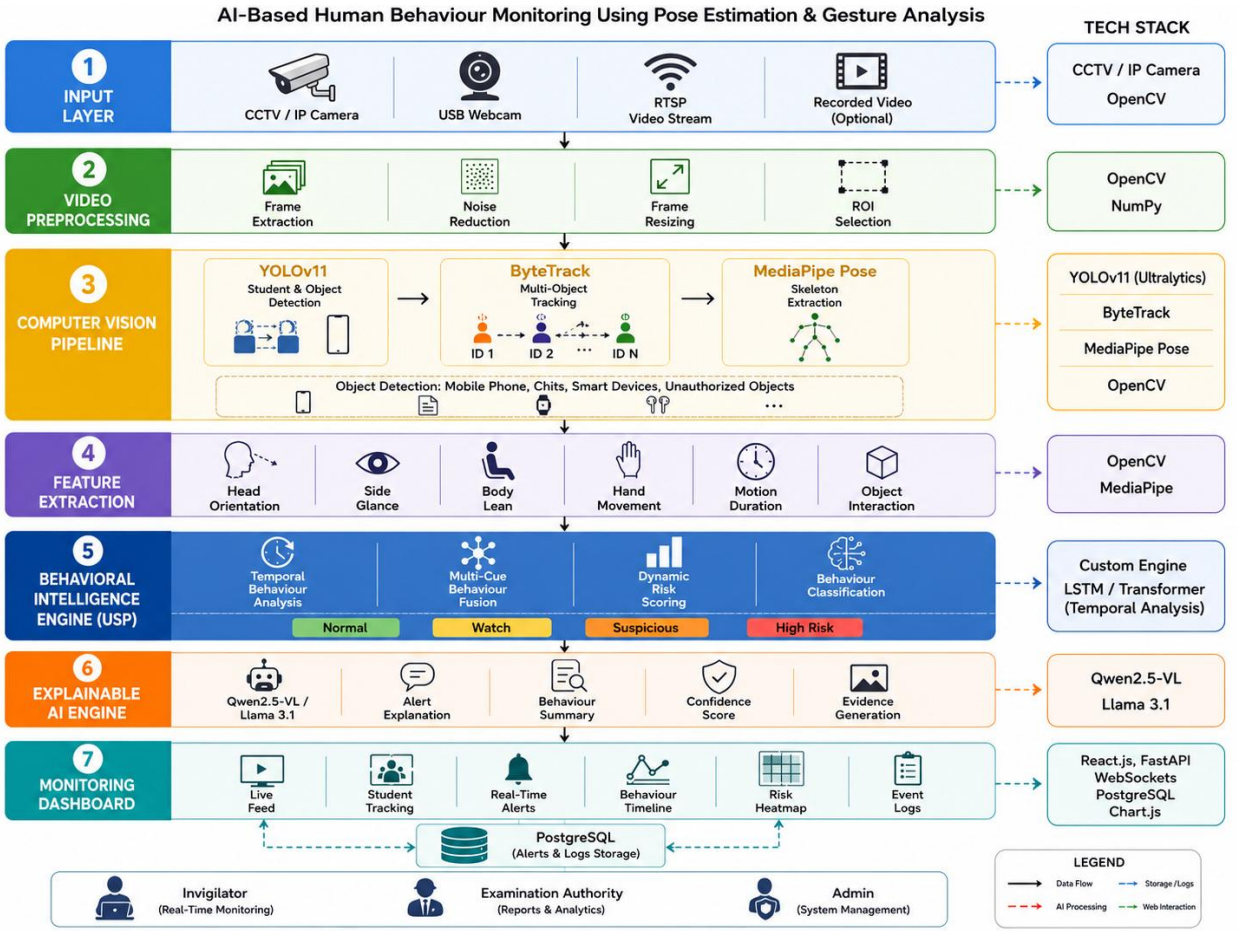
"Black Box" Evidence Pipeline
- A unique end-to-end integration that minimizes manual data entry by automating everything from detection to cloud storage.

TECHNICAL APPROACH

Technical Stack:

- Python
- React.js
- Tailwind CSS
- FastAPI
- OpenCV
- YOLOv11 (Ultralytics)
- MediaPipe Pose
- ByteTrack
- Behavioral Intelligence Engine *(Custom)*
- Temporal Behavior Analysis
- Qwen2.5-VL
- PostgreSQL
- Docker
- Git & GitHub

Methodology:



FEASIBILITY AND VIABILITY

FEASIBLITY:

Privacy-First Architecture:
Uses pose estimation instead of facial recognition, ensuring privacy and data protection.

Behavioral Intelligence Engine:
A multi-cue risk engine analyzes behavioral patterns over time to reduce false positives and detect anomalies.

Real-Time AI Processing:
Optimized YOLOv11, ByteTrack, and MediaPipe models enable efficient, real-time monitoring and inference.

Scalable & Modular Design:
Independent AI modules allow easy integration and scalability from a single classroom to multiple exam halls.

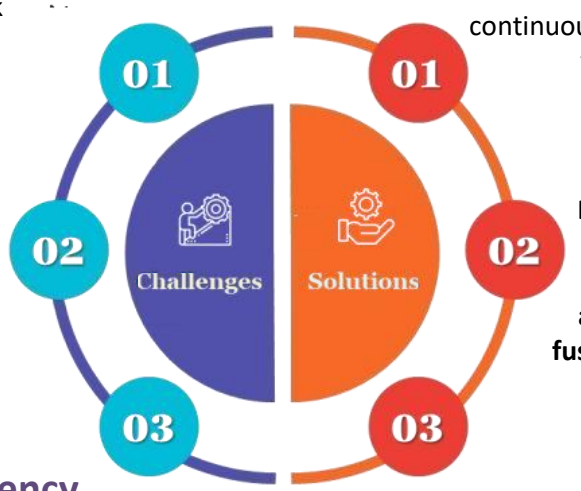
CHALLENGES & SOLUTIONS

Occlusion & Crowded Seating
Students may partially block each other, reducing detection accuracy.

Solution:
Multi-Object Tracking (ByteTrack) maintains continuous tracking even during temporary occlusions..

False Positives
Normal movements may be incorrectly classified as suspicious.

Solution:
Behavioral Intelligence Engine performs **temporal behavior analysis** and **multi-cue fusion** before generating alerts.



Privacy & Transparency
Continuous surveillance raises ethical concerns.

Solution:
No facial recognition is used. **Explainable AI** provides evidence-backed, human-readable alerts, ensuring transparency and accountability.

ROBUSTNESS TESTING & MODEL EVALUATION

COMPUTATIONAL ANALYSIS

Model Complexity & Parameters

- SBAR models range from efficient architectures like BlockGCN (1.3M parameters) to heavier models like MST-GCN (12M), with YOLOv8n-pose requiring 9.2B FLOPs.

Inference Speed & Latency

- PEAR achieves 100+ FPS on standard hardware, while YOLOv8-based systems report 12.94 ms/frame, dropping to 10 FPS on embedded Nvidia Jetson Nano systems.

TinyML & Optimization

- Post-Training Quantization (PTQ) can reduce model size by 4× and improve CPU latency by up to 4× for edge deployment.

MODEL EVALUATION

Core Accuracy Metrics

- GCN-Former achieves high top-1 accuracy on NTU RGB+D 60; other key metrics include Mean Average Precision (mAP) and F1-Score for imbalanced data.

Pose-Specific Metrics

- PCK measures predictions within error thresholds, OKS measures keypoint similarity relative to scale, and MPJPE measures average 3D joint distance in centimeters.

Qualitative Assessments

- Beyond raw data, expert feedback using Likert scales (e.g., from proctors) validates the real-world effectiveness of the models.

EDGE CASE ANALYSIS

Occlusion & Crowding

- Top-down estimation methods are preferred for multi-person scenarios because they isolate individuals before estimating poses, unlike bottom-up methods.

Semantic Ambiguity

- Systems use dynamic thresholding to differentiate between "noisy" data and natural behaviors, such as scratching or yawning, to reduce false positives.

OOD Generalization

- The HumanArt dataset acts as a "stress test" for Out-of-Distribution robustness across non-photorealistic domains like cartoons and historical artwork.

MODEL BENCHMARKS

State-of-the-Art (SOTA) Methods

#	Dataset	Metric	Leading Models	Performance Example
1	NTU RGB+D 60	Top-1 Acc (X-Sub)	GCN-Former, BlockGCN, HD-GCN	93.5% (GCN-Former)
2	NTU RGB+D 120	Top-1 Acc (X-Sub)	GCN-Former, BlockGCN, CTR-GCN	90.7% (GCN-Former)
3	NW-UCLA	Top-1 Acc	FG-STFormer, CTR-GCN	97.0% (FG-STFormer)
4	MSCOCO	mAP	ViTPose, TransPose, TokenPose	> 80 mAP
5	MPII	PCKh@0.5	Nie et al., FPD	92.4% (Nie et al.)
6	LSP	PCK@0.2	Peng et al., FPD	94.5% (Peng et al.)

RESEARCH AND REFERENCES

Title	Approach	Dataset Used	Results / Accuracy
ByteTrack: Multi-Object Tracking by Associating Every Detection Box (ECCV 2022)	Associates every detection box (including low-confidence detections) to maintain robust tracking under occlusion.	MOT17, MOT20, HiEve, BDD100K	80.3 MOTA, 77.3 IDF1, 63.1 HOTA, 30 FPS (arXiv)
Multiple Instance Learning for Cheating Detection and Localization in Online Examinations (2024)	Combines body posture, eye gaze, head pose, facial features, and spatio-temporal graph learning for cheating detection.	Online Exam Proctoring (OEP), UCF-Crime, ShanghaiTech	87.58% Frame-level AUC on OEP dataset (arXiv)
Real-Time AI-Driven Exam Cheating Detection Using YOLO and Pose Estimation (2025)	Multi-modal framework integrating YOLO object detection with human pose estimation for real-time invigilation.	Custom annotated examination video dataset	Precision: 92%, Recall: 88%, F1-Score: 90% (IRE Journals)
Cheating Recognition in Examination Halls Based on Improved YOLOv8 (2025)	Enhanced YOLOv8 with hierarchical detection for recognizing cheating behaviors and prohibited objects.	Custom examination hall dataset	Improved detection accuracy over baseline YOLOv8 (Springer Link)
A Two-Stage, Object-Centric Deep Learning Framework for Robust Exam Cheating Detection (2026)	YOLO-based student detection followed by behavior classification using RexNet-150 for ethical and scalable monitoring.	Combined dataset from 10 public sources (273,897 samples)	95% Accuracy, 96% Precision, 94% Recall, 95% F1-Score (arXiv)